

Figure legend

Figure 1. Reported EMC junctions on the retained-RG axis, and the type-2 seam. Three panels, drawn by `emc_fusion_frame_figure.py` and committed as `emc-fusion-frame-fig1.png` and `emc-fusion-frame-fig1.pdf`.

Panel A. Bars are drawn to residue scale on the shared axis. Each thin vertical tick marks one RG dipeptide, at the 1-based position of its arginine (`emc-fet-frame-and-composition.json`, `composition.rg_content.<protein>.rg_positions`); the count printed at the right of each wild-type bar is that protein's total. Dashed rectangles mark operationally defined RGG boxes — maximal merged runs of 60-residue windows each containing at least six RG dipeptides, trimmed to start at the first RG the box contains and to end two residues past its last RG-start (`emc_fet_idr_census.py`, `RGG_WINDOW = 60`, `RGG_MIN_RG_IN_WINDOW = 6`); the spans drawn are `rgg_boxes_operational` in `emc-fet-construct-designs.json`. This finder returns two boxes on EWSR1 and one on TAF15; the source paper names three RGG-rich domains in EWSR1, and the box definition was not tuned to reproduce that count. NR4A3 has no box because the finder returns none for it. C166 is a printed text label placed above the NR4A3 bar with a single vertical marker drawn on that bar at residue 166; it identifies that residue position and nothing more, and no functional property is asserted for it here. Below the dotted rule, each fusion row shows the retained 5' segment ending at a heavy end-cap at its breakpoint, carrying only the RG ticks at or before that breakpoint; the right-hand text gives the retained residue range and the retained-of-total RG count. Panel A carries four weights of vertical black line — RG tick, dashed box outline, breakpoint end-cap and the C166 marker — and the figure prints no key of its own, so the reader is directed to this caption for all four.

Panel B. The retained-RG fraction axis, carrying the fusion positions of Table 3, the hatched band labelled "position on this axis not determinable" for the construct whose reintroduced RGG-rich domain the source does not identify, and the 0.000 to 0.267 interval spanned by the three reported EWSR1::ATF1 breakpoints. That interval describes reported breakpoints; it is not a measured recruitment range.

Panel C. The type-2 seam, drawn as one nucleotide donated by EWSR1 followed by 176 nt of NR4A3 untranslated sequence: 174 nt of NR4A3 exon 2, the block labelled untranslated in NR4A3, and the first 2 nt of NR4A3 exon 3, annotated separately in the panel. Those 177 nt are the extension of 59 codons carrying no internal stop codon. The earlier caption counted only the exon 2 block and the donated nucleotide, which does not reach 177. The panel does not print the words "type 2": it draws the EWSR1 exon 7 to NR4A3 exon 2 junction, which is the only reported EMC junction using the exon 2 acceptor, and it should be read as that junction and not as a general statement about the acceptor.

What the figure shows, and what it does not. The three panels carry the reported junctions, the RG content each retains, the retained-RG fraction axis and the type-2 seam. The frame rule as a rule over all donor and

acceptor pairs, the complete phase-1 donor set, the symmetric prefix sweep and its 0.039 margin, and the TAF15 zero-RG margin are results of sections 3.2, 3.4 and 3.5 and appear in no panel.